Lead *DeFi* Software Engineer — *Trading* & *Blockchain* Infrastructure

Location: Remote / Poland (International Team)
Employment Type: Full-time / Part-time / Contract
Department: *Blockchain* & Financial Engineering
Reports To: Head of Engineering / CTO
Level: Lead Engineer

About the Company

We are preparing to establish an independent software development company in Poland, focused on building secure, scalable fintech and *blockchain* applications.

Our planned products and technical initiatives include:

- **Decentralized Exchanges (*DEX*):** *Trading* platforms, *token swap*s, liquidity management, and on-chain financial applications.
- ***DeFi* Infrastructure:** Lending, borrowing, staking, collateral management, and yield-related financial products.
- **Digital Asset Platforms:** *Cryptocurrency wallet*s, asset management, transaction processing, and *blockchain* integrations.
- **Fintech & Payments:** Stable*coin* payments, financial APIs, transaction settlement, and digital asset services.
- ***Blockchain* Infrastructure:** *Smart contract* integrations, transaction in*dex*ing, cross-chain services, and reliable distributed backend systems.

We aim to connect modern financial technology with decentralized infrastructure while maintaining s*trong* standards for security, performance, and reliability.

We are building an international engineering team and looking for experienced engineers who can contribute to important architectural decisions while remaining actively involved in development.

About the Role

We are looking for a **Lead *DeFi* Software Engineer — *Trading* & *Blockchain* Infrastructure** to design, develop, and lead engineering work across decentralized *trading* applications and *blockchain*-connected financial systems.

This is a hands-on technical leadership position combining *DeFi* protocol knowledge, backend architecture, *smart contract* integration, and *trading* infrastructure development.

You will help build reliable systems supporting *trading* execution, *blockchain* transactions, liquidity management, financial data processing, and secure interactions with decentralized protocols.

The ideal candidate has s*tron*g experience with ***Solidity*, TypeScript/Node.js, and modern backend engineering**, along with practical knowledge of *Ethereum*, *EVM* networks, and *DeFi* protocols.

Experience with *Rust*, Go, low-latency *trading* systems, and *blockchain* in*dex*ing is particularly valuable.

What You'll Do

DeFi Protocol & *Smart Contract* Engineering

- Design, develop, integrate, and maintain *smart contract*s supporting *DeFi* applications.
- Work with *Ethereum* and *EVM*-compatible *blockchain* networks.
- Develop or integrate *trading*, lending, staking, and liquidity-related contract functionality.
- Collaborate on *Solidity* architecture, contract security, and upgrade strategies.
- Implement *smart contract* interactions using viem, *eth*ers.js, or comparable libraries.
- Develop reliable handling of *token* approvals, transfers, contract calls, and transaction confirmations.
- Support integration with established protocols such as *Uniswap*, *Aave*, Curve, or similar platforms.
- Review protocol assumptions, economic edge cases, and security risks.

Trading & Execution Infrastructure

- Design backend services for decentralized *trading* applications.
- Build transaction execution workflows for *swap*s, orders, liquidity operations, and settlement.
- Integrate decentralized exchange aggregators and *trading* protocols.
- Develop services for price quotations, routing, slippage protection, and transaction monitoring.
- Support real-time market data processing and *trading*-related APIs.

- Optimize transaction submission, confirmation handling, and execution reliability.
- Implement appropriate retries, idempotency, and failure recovery mechanisms.
- Collaborate on risk controls, balance validation, and transaction lifecycle management.

Backend & Distributed Systems

- Design and develop scalable backend services using Node.js, TypeScript, Go, *Rust*, or comparable technologies.
- Build secure REST, GraphQL, and WebSocket APIs.
- Implement event-driven services for *blockchain* data processing and *trading* activity.
- Work with PostgreSQL, Redis, and message brokers such as Kafka or RabbitMQ.
- Develop *blockchain* in*dex*ing, event monitoring, and transaction tracking services.
- Design reliable communication between off-chain services and on-chain contracts.
- Optimize backend latency, throughput, concurrency, and resource usage.
- Maintain consistency between financial application state and *blockchain* transaction state.

Blockchain Infrastructure & Integration

- Integrate *blockchain* RPC providers, nodes, in*dex*ers, and data services.
- Develop infrastructure supporting multiple *EVM*-compatible networks.
- Build services for *token* balances, transaction histories, contract events, and asset movements.
- Support *wallet* integrations, transaction signing workflows, and *blockchain* confirmations.
- Handle chain reorganizations, failed transactions, RPC outages, and network congestion.
- Work with infrastructure engineers to improve node connectivity and system resilience.
- Evaluate cross-chain integrations and interoperability protocols where required.
- Maintain secure handling of sensitive transaction and application data.

Security, Testing & Reliability

- Follow secure engineering practices for systems processing financial transactions.
- Review *smart contract*s and backend services for security vulnerabilities.
- Develop automated unit, integration, fuzz, and invariant tests where appropriate.
- Use Foundry, Hardhat, or comparable tools for *smart contract* testing.
- Identify risks involving access controls, reentrancy, external calls, oracle data, and transaction ordering.
- Implement monitoring, logging, tracing, and operational alerts.
- Participate in security reviews, audits, incident investigations, and remediation.
- Improve deployment reliability using CI/CD and automated validation.

Technical Leadership

- Lead important *DeFi* and *trading* infrastructure engineering initiatives.
- Own architectural decisions for selected backend and *blockchain* components.
- Remain actively involved in coding, debugging, and code reviews.
- Mentor engineers and support technical knowledge sharing.
- Collaborate with frontend, *blockchain*, security, DevOps, and product teams.
- Translate product requirements into practical technical designs.
- Establish engineering standards for security, testing, maintainability, and performance.
- Help prioritize technical improvements alongside product delivery.

Required Skills & Experience

- 6+ years of professional software engineering experience.
- 3+ years of relevant *blockchain*, *DeFi*, or financial infrastructure engineering experience.
- S*trong* hands-on experience with TypeScript/Node.js, *Solidity*, or comparable technologies used in production *DeFi* systems.
- Practical experience integrating *Ethereum* or *EVM*-compatible *blockchain* networks.
- Understanding of *DeFi* protocols, decentralized exchanges, *token* standards, and *smart contract* interactions.
- Experience designing and maintaining production backend services and APIs.
- S*trong* knowledge of distributed systems, asynchronous processing, and transaction reliability.
- Experience with PostgreSQL, Redis, or comparable databases and caching systems.
- Familiarity with *blockchain* RPC APIs, event in*dex*ing, transaction signing, and confirmation tracking.
- Understanding of financial transaction workflows, balances, settlement, and failure recovery.
- Experience with automated testing, debugging, and secure software development.
- Familiarity with Git, Docker, CI/CD, and cloud infrastructure.
- Ability to lead engineering initiatives and guide other developers.
- S*trong* problem-solving skills and effective communication in a remote environment.

Nice to Have (Differentiators)

- Experience with *Uniswap* V2/V3/V4, Curve, *Aave*, GMX, or similar *DeFi* protocols.
- Familiarity with automated market makers, liquidity pools, and lending protocols.
- Experience with Foundry, fuzz testing, invariant testing, or Certora.
- Knowledge of MEV, Flashbots, CoW Protocol, or intent-based *trading*.
- Experience developing *DEX* aggregators, *trading* bots, or execution infrastructure.
- S*trong* *Rust* or Go development experience.
- Knowledge of *blockchain* in*dex*ers such as Ponder, Envio, or The Graph.
- Experience with financial risk controls, liquidation workflows, or collateral management.

- Familiarity with cross-chain messaging and transaction execution.
- Experience with *smart contract* auditing and formal verification.
- Knowledge of AWS, Kubernetes, Terraform, and observability tools.
- Contributions to open-source *DeFi* or *blockchain* infrastructure projects.

Education

- Bachelor's or Master's degree in Computer Science, Software Engineering, Mathematics, or a related technical field.
- Equivalent professional experience and demonstrated expertise in production *DeFi* engineering are equally valued.
- Open-source contributions and independently verifiable engineering projects are appreciated.

What Success Looks Like

First 3 Months

- Understand the planned *DeFi* products, *trading* workflows, and technical architecture.
- Evaluate core *smart contract* integrations and backend infrastructure requirements.
- Identify important security, reliability, and performance risks.
- Deliver initial *trading* or *blockchain* integration components.
- Establish practical development, testing, and code review standards.

Within 6 Months

- Take ownership of major *DeFi* backend or *trading* infrastructure components.
- Deliver reliable transaction execution and *blockchain* integration workflows.
- Improve performance, monitoring, and failure recovery.
- Guide engineers through complex technical implementation challenges.
- Support security reviews and production readiness.

Long-Term

- Help establish scalable infrastructure supporting multiple *DeFi* and digital asset products.
- Improve transaction reliability, security, and system performance.
- Support expansion into additional *blockchain* networks and *trading* integrations.
- Establish reusable engineering components and architectural standards.
- Provide technical leadership while remaining involved in hands-on development.

Why Join Us?

- Help build fintech and *DeFi* infrastructure at an early-stage independent software company.
- Work on challenging financial engineering and *blockchain* integration problems.
- Contribute to decentralized exchanges, digital asset platforms, and payment infrastructure.
- Take meaningful ownership of technical architecture and development decisions.
- Work with an international engineering team.
- Enjoy flexible remote collaboration and employment arrangements.
- Combine hands-on software development with technical leadership responsibilities.

Ideal Candidate

We are looking for an experienced, hands-on engineer who understands both *blockchain* applications and the backend systems required to operate them reliably.

You should be comfortable integrating *DeFi* protocols, building transaction execution services, designing distributed backend components, and investigating complex financial or *blockchain*-related failures.

You should have s*trong* security awareness and understand the importance of correctness when developing systems that process financial transactions.

We value engineers who can lead technical decisions, mentor other developers, and contribute directly to production code.

The ideal Lead DeFi Software Engineer combines DeFi protocol knowledge, strong backend engineering skills, transaction execution expertise, and practical technical leadership.